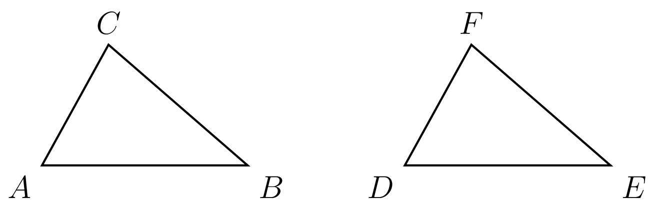
Triangle Congruence: Solving for Corresponding Parts

High School Geometry

Directions

- A congruence statement is read **in order**: in $\triangle ABC \cong \triangle DEF$ the first letters correspond ($A \leftrightarrow D$), the second ($B \leftrightarrow E$), the third ($C \leftrightarrow F$). Corresponding sides are named by corresponding letter pairs: $\overline{AB} \leftrightarrow \overline{DE}$.
- By **CPCTC**, corresponding parts of congruent triangles are congruent, so the two matched measures are *equal*. Write that equation first, then solve it.
- Every length below is in the unit stated in its problem, and every angle measure is in degrees. Show the equation you solved.



How a congruence statement is read on this sheet: matching positions name matching parts. **No values shown** — use the measures given in each problem.

1. $\triangle ABC \cong \triangle DEF$. The side $\overline{AB}$ measures $3x+2$ centimetres and the side $\overline{DE}$ measures 17 centimetres. Write the equation CPCTC gives you, and solve for x.

Answer: ________________

2. $\triangle PQR \cong \triangle STU$. The angle $\angle P$ measures $(4y-10)$ degrees and the angle $\angle S$ measures 50 degrees. Solve for y.

Answer: ________________

3. $\triangle JKL \cong \triangle MNP$. The side $\overline{JK}$ measures $2x + 7$ inches and the corresponding side $\overline{MN}$ measures $4x - 9$ inches. Solve for x.

Answer: ________________

4. $\triangle ABC \cong \triangle DEF$. In $\triangle ABC$, the angle $\angle A$ measures 52 degrees and the angle $\angle B$ measures 61 degrees. Find the measure of $\angle F$ in degrees. (Which angle of $\triangle ABC$ corresponds to $\angle F$?)

Answer: ________________

5. $\triangle RST \cong \triangle XYZ$. The angle $\angle R$ measures $(5x - 3)$ degrees and the angle $\angle X$ measures $(2x + 12)$ degrees.

(a) Solve for x.

Answer: ________________

(b) Find the measure of $\angle R$.

Answer: ________________ degrees

6. Bridge truss. Two triangular panels of a truss are built congruent: $\triangle ABC \cong \triangle DEF$. The steel member $\overline{AC}$ measures $4x + 1$ metres and the corresponding member $\overline{DF}$ measures $2x + 13$ metres.

(a) Solve for x.

Answer: ________________

(b) How long is member $\overline{AC}$?

Answer: ________________ m

7. Points A, B, C and D are arranged so that $\overline{AB} \cong \overline{CD}$ and $\angle BAC \cong \angle DCA$. The segment $\overline{AC}$ belongs to both $\triangle BAC$ and $\triangle DCA$.

(a) Prove $\triangle BAC \cong \triangle DCA$. Write a two-column proof; the third pair of parts you need is the shared side.

(b) Using that congruence, $\overline{BC}$ measures $3x - 4$ centimetres and its corresponding side $\overline{DA}$ measures 20 centimetres. Solve for x.

Answer: ________________

8. Find and fix the mistake. Kim is given $\triangle ABC \cong \triangle DFE$ and writes "$\overline{AB}$ corresponds to $\overline{DE}$".

(a) Explain why Kim's correspondence is wrong, and name the side that really corresponds to $\overline{AB}$.

Answer: ________________

(b) The side $\overline{AB}$ measures $2x + 5$ centimetres and the side $\overline{DF}$ measures 31 centimetres. Write the correct equation and solve for x.

Answer: ________________